

Simplify ratios

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Simplify 36:48. Copy or complete the first step.

2. Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

3. There are 8 red and 12 blue counters. Write red to blue as a ratio.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Simplify 36:48. Copy or complete the first step.

Answer 3:4.

2. Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

Answer 30 ml.

3. There are 8 red and 12 blue counters. Write red to blue as a ratio.

Answer 8:12, which simplifies to 2:3.

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

Simplify ratios

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Simplify 36:48.

6. Check this result using an inverse, estimate or known fact: There are 8 red and 12 blue counters. Write red to blue as a ratio.

2. Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

7. Prove or explain your reasoning: Explain the ratio 5:2 in a sentence.

3. There are 8 red and 12 blue counters. Write red to blue as a ratio.

8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.

4. Solve this independently and show every stage: Simplify 36:48.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

10. Write and solve a similar question to: There are 8 red and 12 blue counters. Write red to blue as a ratio.

Teacher answers and checking prompts

1. Simplify 36:48.
Answer 3:4.
2. Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer 30 ml.
3. There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer 8:12, which simplifies to 2:3.
4. Solve this independently and show every stage: Simplify 36:48.
Answer 3:4.
5. Use a second method or representation for: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer Expected result: 30 ml. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer Expected result: 8:12, which simplifies to 2:3. A valid check must be shown.
7. Prove or explain your reasoning: Explain the ratio 5:2 in a sentence.
Answer For every 5 of the first quantity, there are 2 of the second. The explanation must show why the method works.
8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Simplify ratios

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Explain the ratio 5:2 in a sentence.

6. Create a new example that tests the same idea as: There are 8 red and 12 blue counters. Write red to blue as a ratio.

2. Prove the answer to this question, rather than only calculating it: Simplify 36:48.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 8:12, which simplifies to 2:3.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Explain the ratio 5:2 in a sentence.
Answer For every 5 of the first quantity, there are 2 of the second. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Simplify 36:48.
Answer Expected result: 3:4. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer Expected result: 30 ml. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 8:12, which simplifies to 2:3.
Answer One valid reconstruction is: There are 8 red and 12 blue counters. Write red to blue as a ratio. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
6. Create a new example that tests the same idea as: There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.